ICS 4111: Embedded Systems & IoT

Practical Exercise 3: Embedded System Design

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Experiment 1

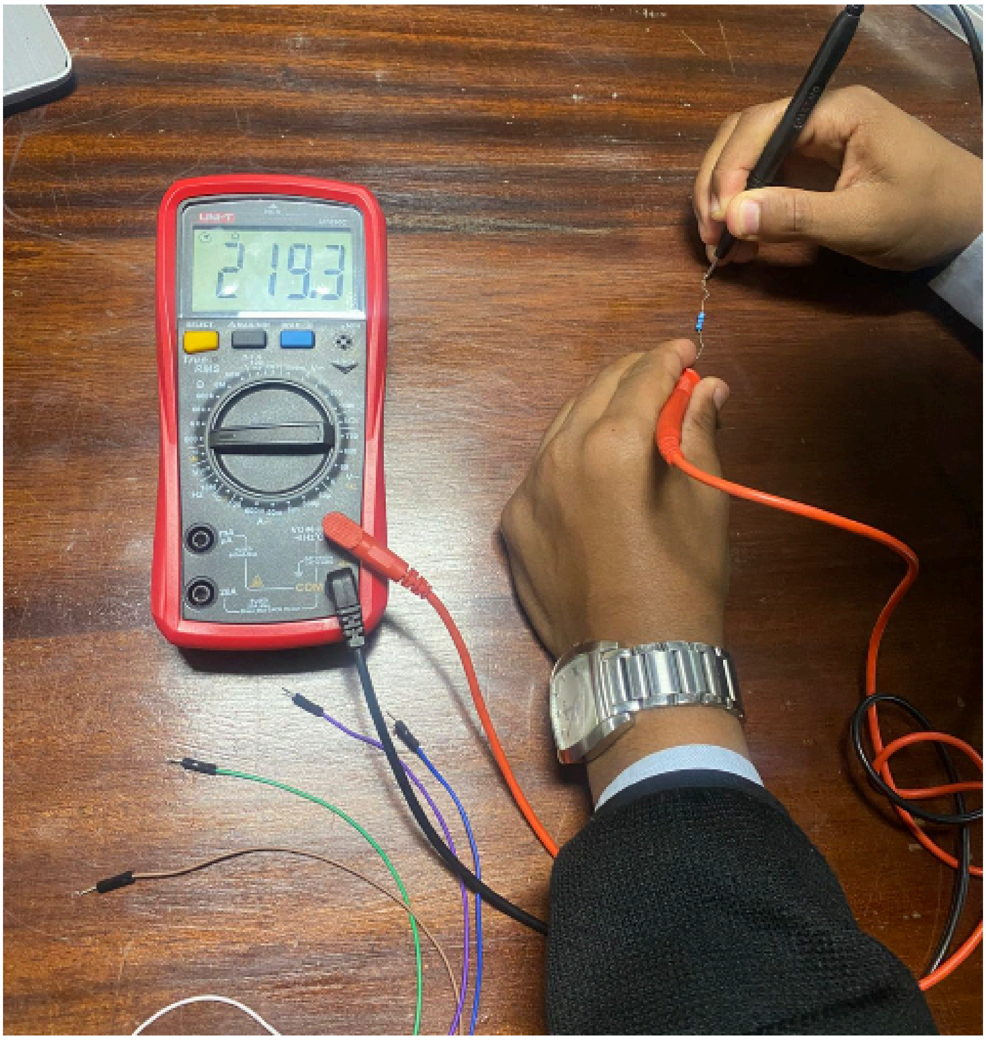
Question 1 – Resistor Identification (Colour Code)

Figure 1: Selected resistor and colour code

Resistance Value- **200 Ohms.**

Question 2 – Resistance Measurement (Multimeter)

Figure 2: Multimeter measuring resistor value

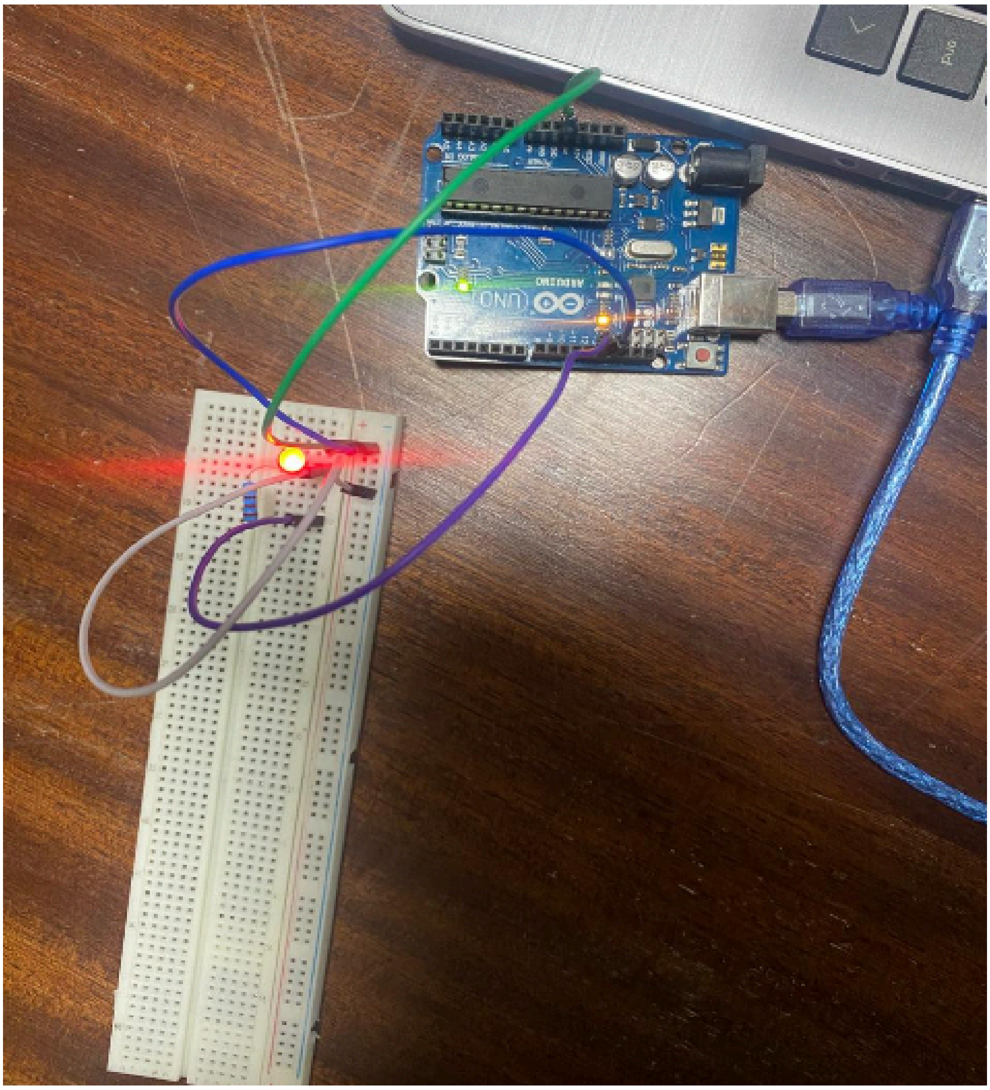


Measured Resistance: **219.3 Ohms.**

Experiment 2

Question 1 – Circuit Setup

Figure 3: Physical circuit implementation — Arduino UNO + LED



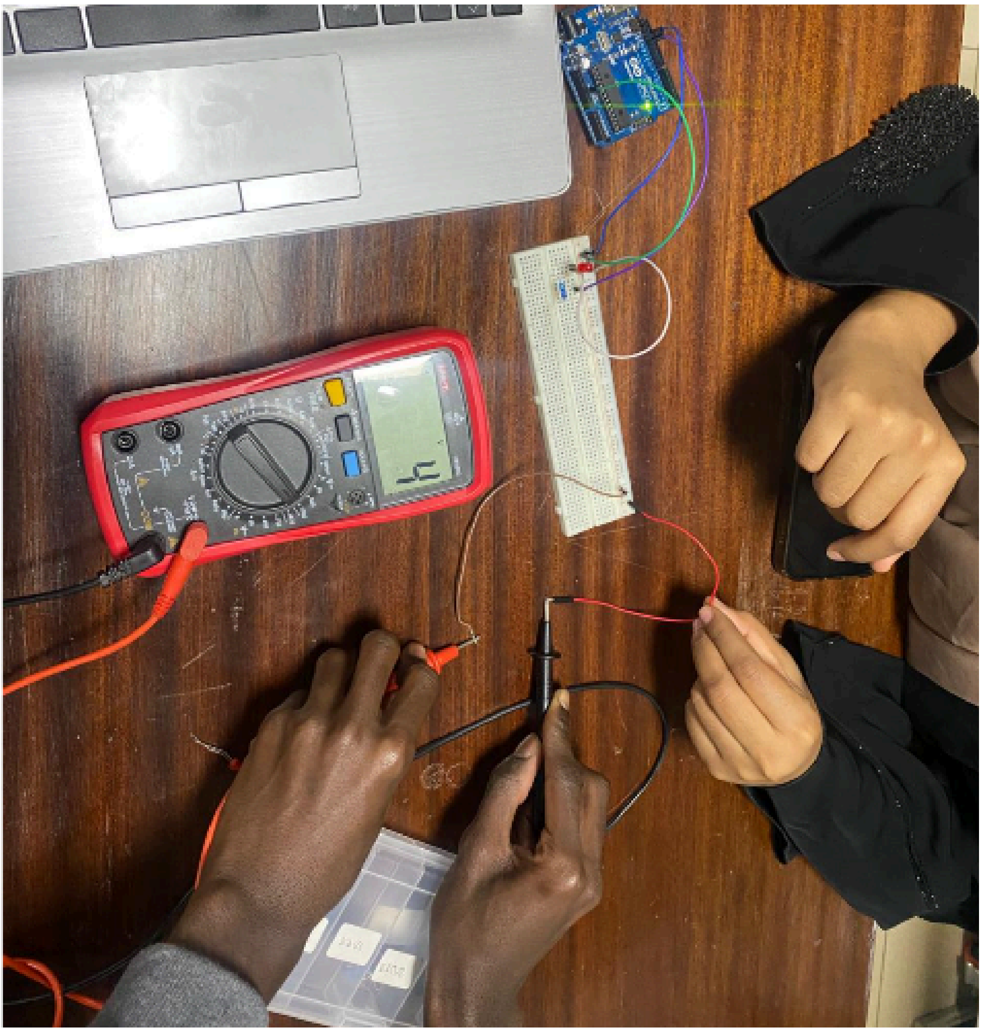
Question 2 - Code

```
void setup() {
  pinMode(13, OUTPUT);
}

void loop() {
  digitalWrite(6, HIGH);   // LED on
  delay(1000);
  digitalWrite(6, LOW);    // LED off
  delay(1000);
}
```

Question 3 – Voltage Measurement

Figure 4: Multimeter measuring voltage across the circuit



Multimeter Value: **4V**

Question 4 – PWM Brightness Control (Potentiometer)

Figure 5: LED at 25% brightness

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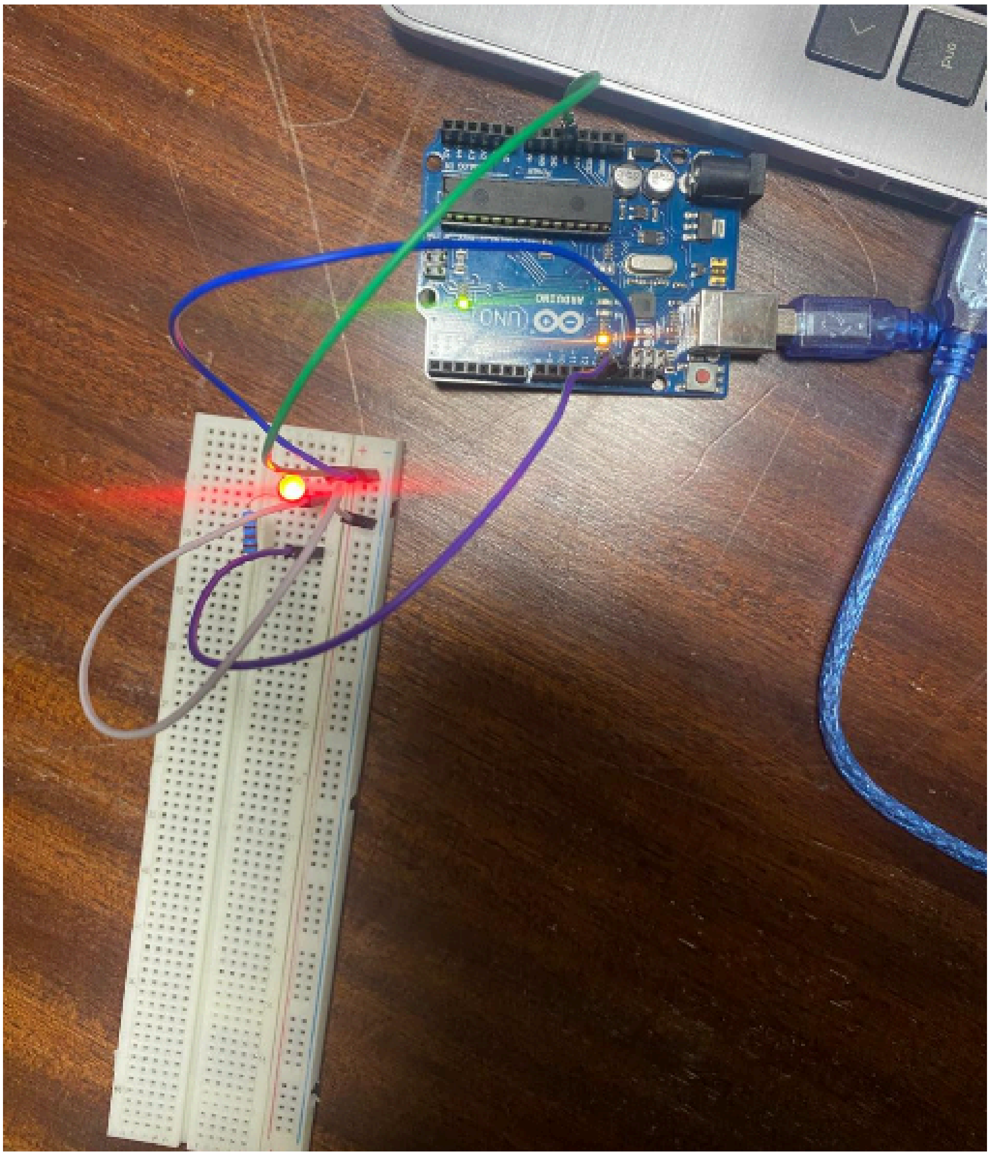


Figure 6: LED at 50% brightness

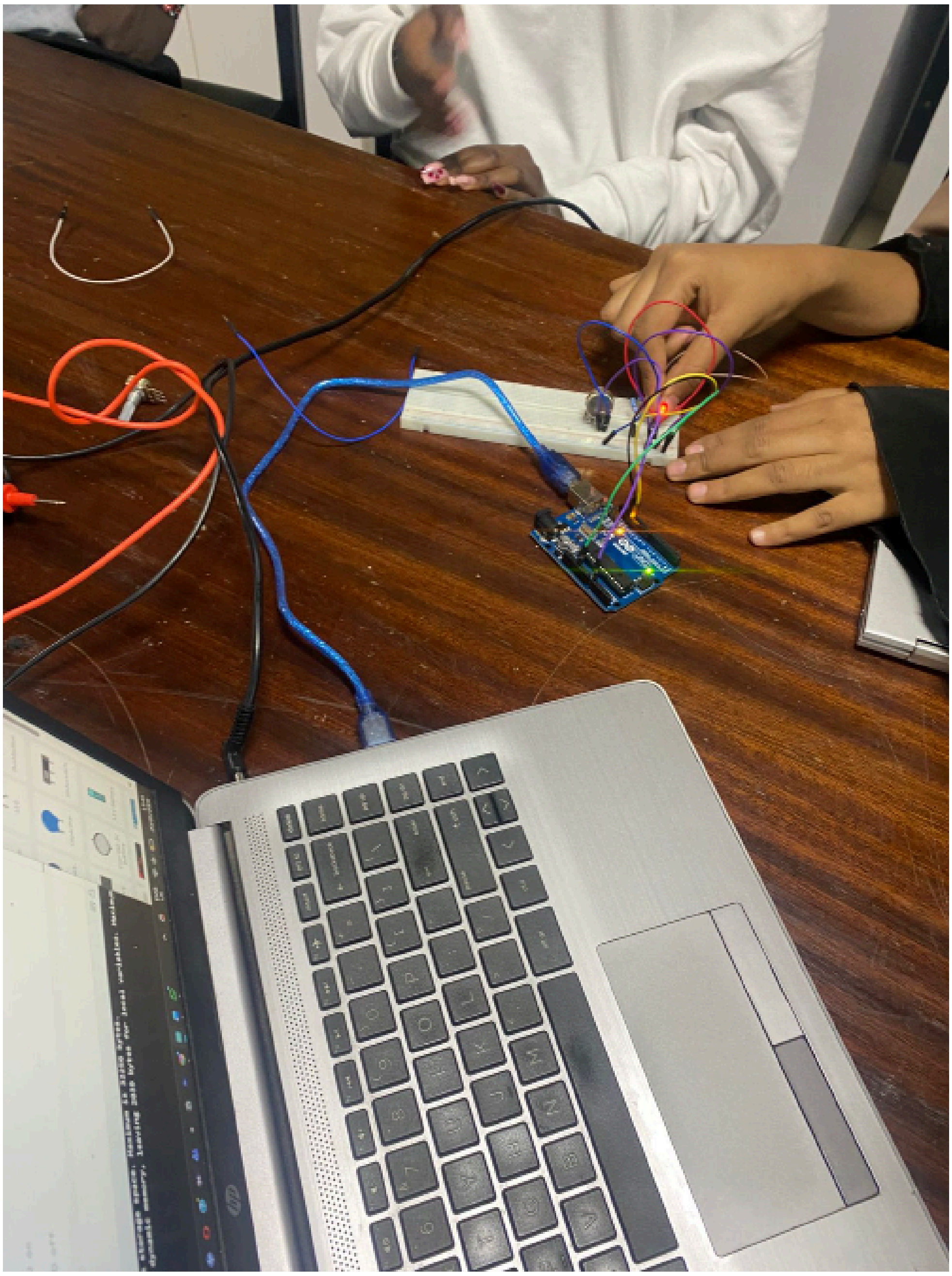
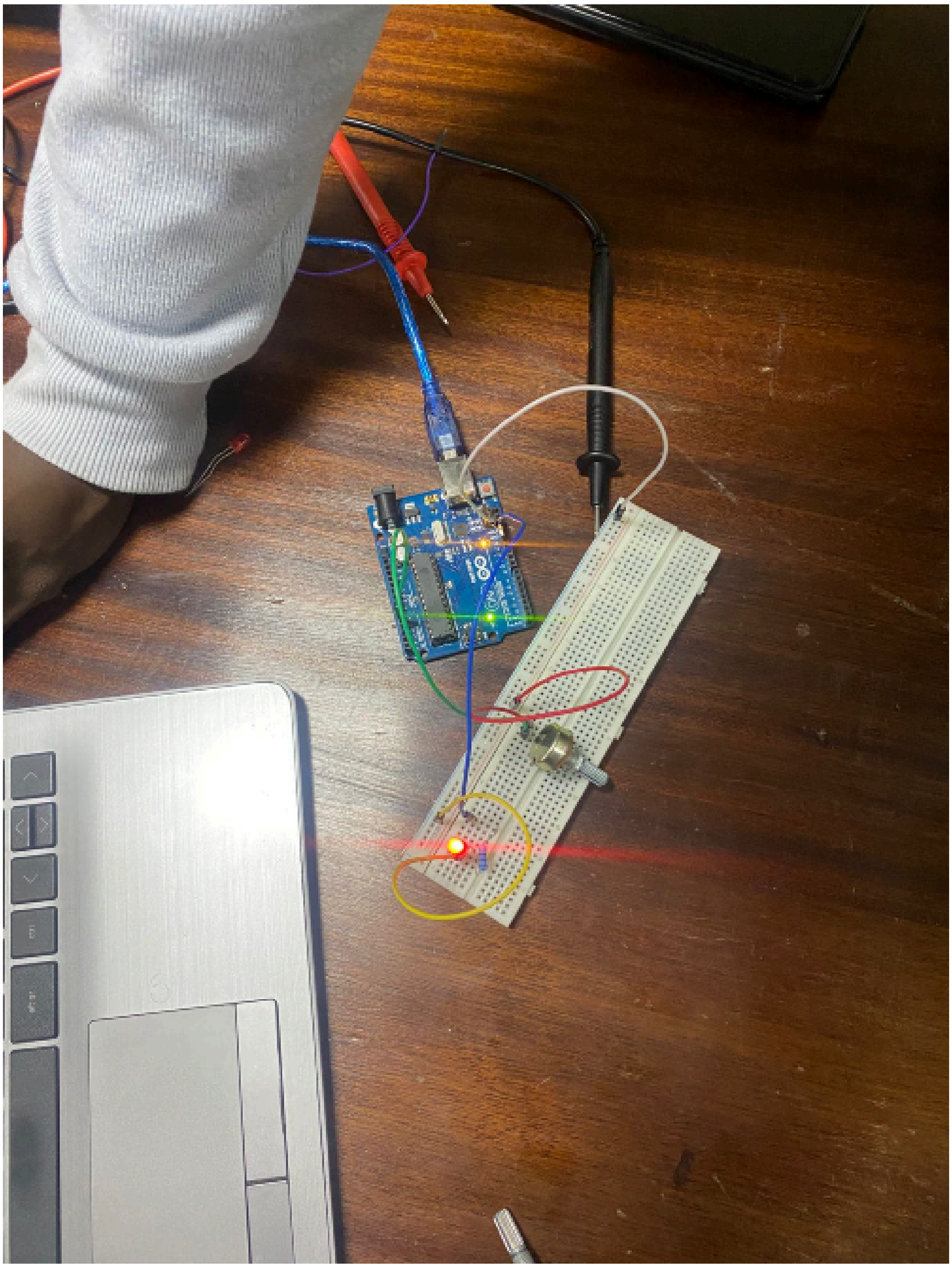


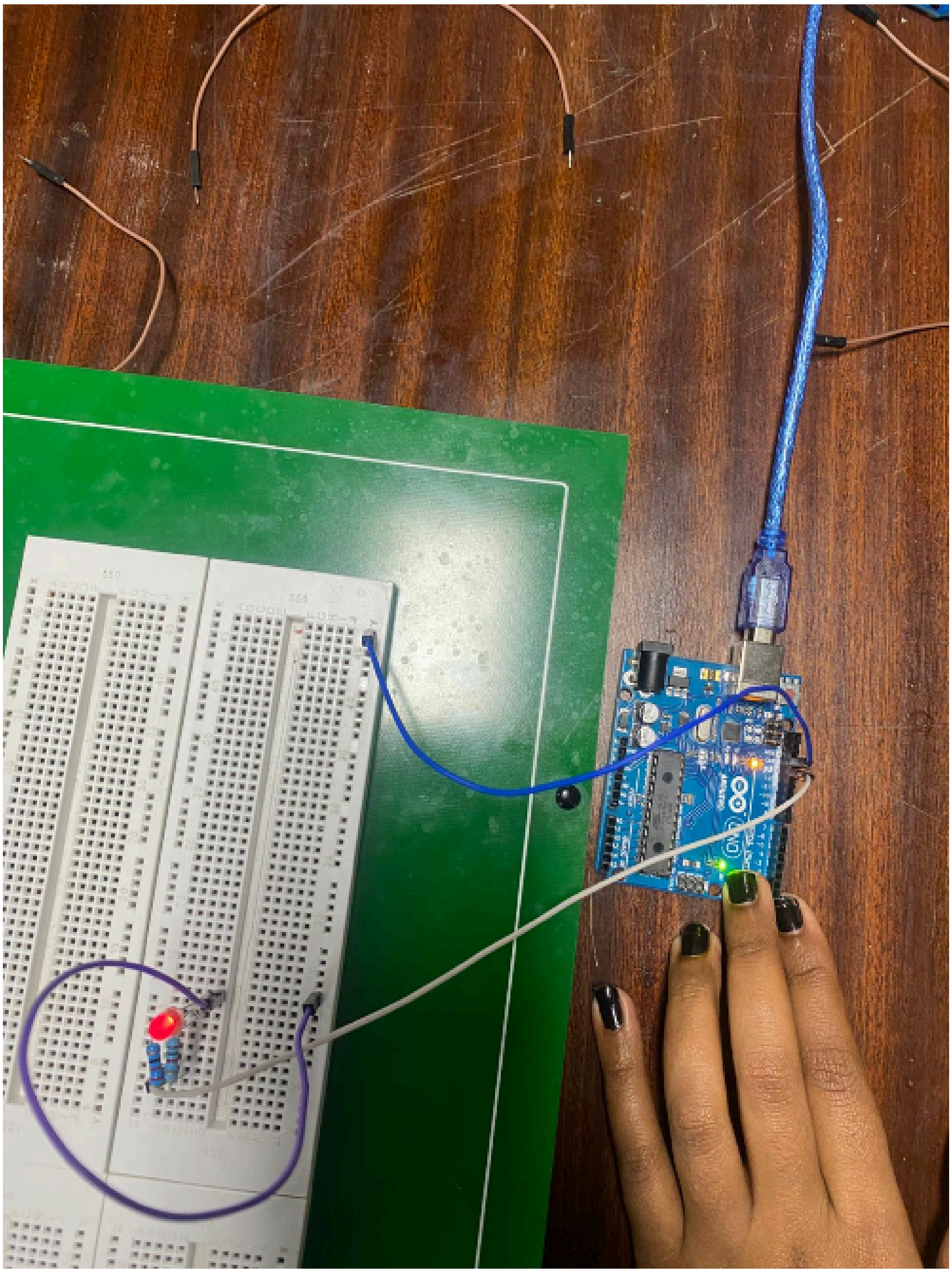
Figure 7: LED at 100% brightness



Experiment 3

Question 1 – Series Resistor Circuit

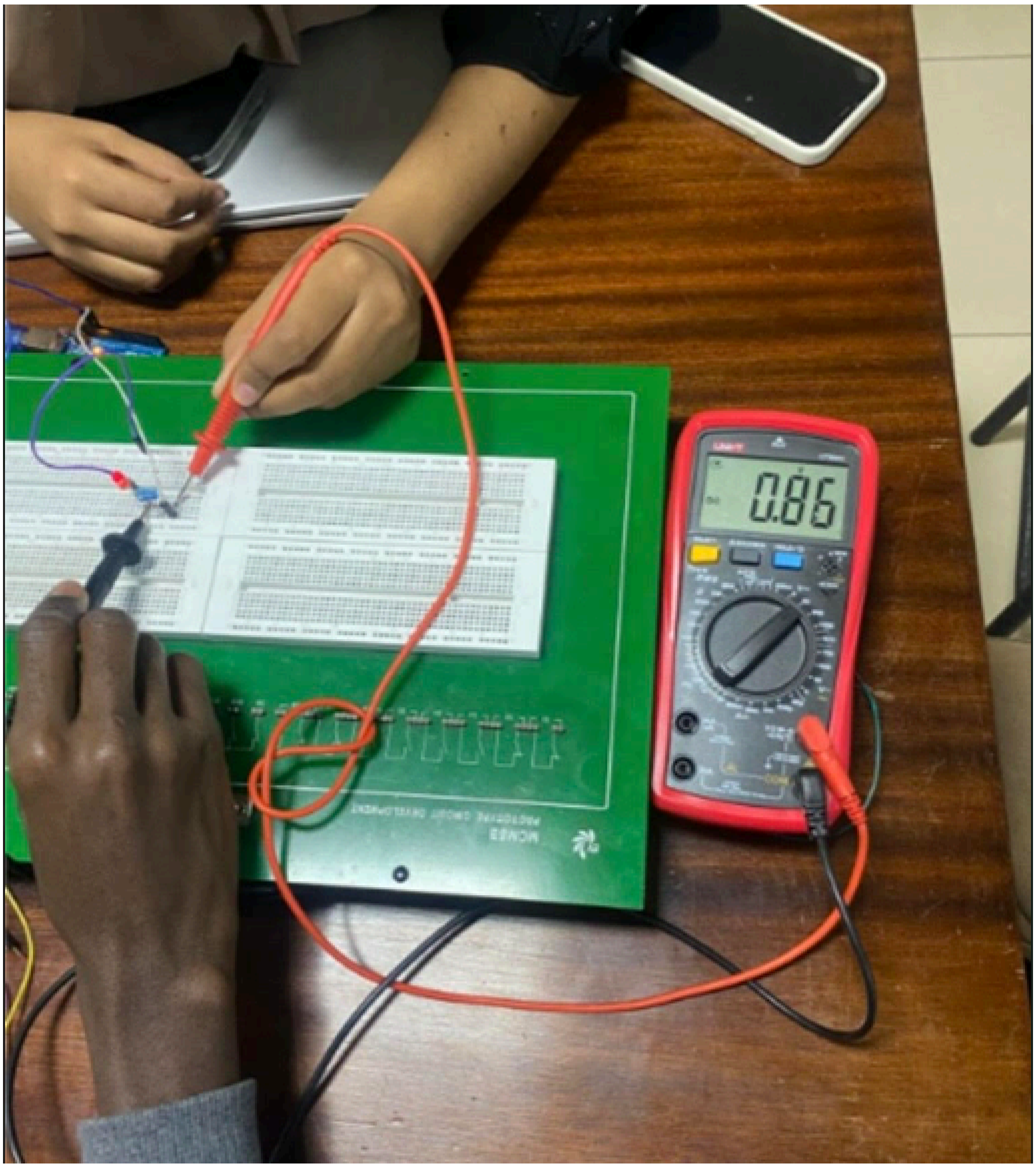
Figure 8: Physical circuit resistors connected in series



Question 2 - Code

```
void setup() {
  pinMode(8, OUTPUT);
}

void loop() {
  digitalWrite(8, HIGH);   // LED on
  delay(1000);
  digitalWrite(8, LOW);    // LED off
  delay(1000);
}
```

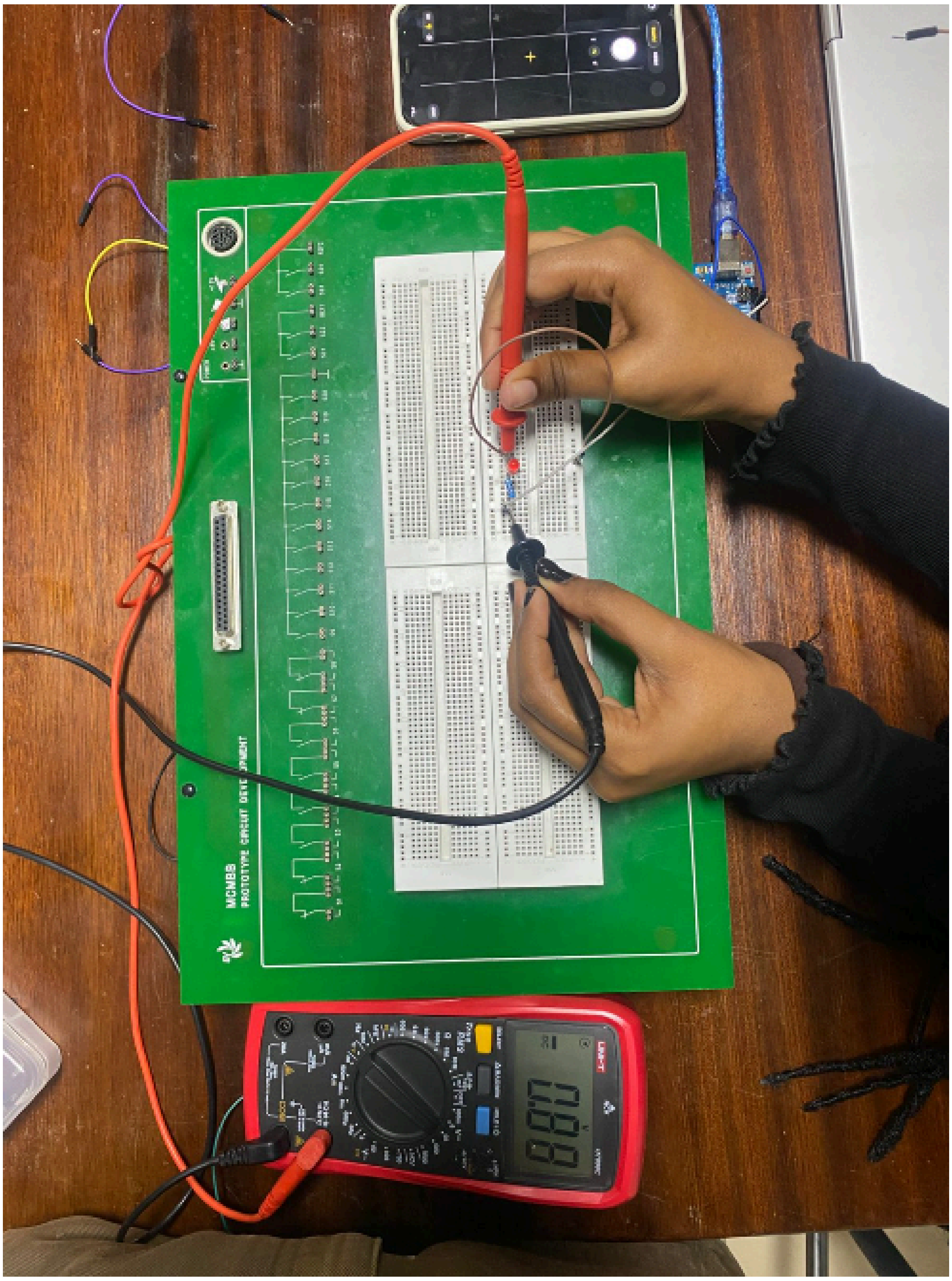


Voltage = 0.86V.

Experiment 4

Question 1 – Parallel Resistor Circuit

Figure 9: Physical circuit — resistors connected in parallel



Question 2 - Code

```
void setup() {
  pinMode(6, OUTPUT);
}

void loop() {
  digitalWrite(6, HIGH);   // LED on
  delay(1000);
  digitalWrite(6, LOW);    // LED off
  delay(1000);
}
```

Voltage = **0.88V.**

Group Member Photo.